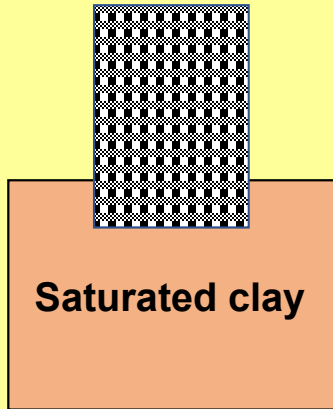
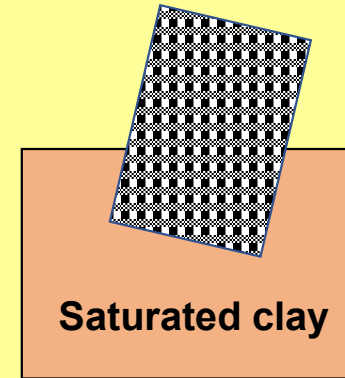


Module 8
Lecture 8.1a Consolidation

A typical field problem



Immediately after construction



Long time after construction

High compressibility of soil induce significant settlement with time

This can result in failure of structures or make it non-serviceable

It is important to assess the compressibility of soil before the construction **(Prevent as far as possible)**

This would help to determine the total settlement and time duration of this settlement

Can take adequate precaution and improve the soil before construction

Undisturbed sample taken from the foundation location for determining compressibility

Compression of soil

Total stress increases in the soil due to the construction of foundation, embankment, or any other structure

The increase in stress causes decrease in soil volume (**compression**)

The soil property that quantifies the amount of compression under a given load is called **compressibility**

Causes of compression are (a) deformation of soil (b) relocation of soil particles (c) expulsion of water/ air from voids

Compression of soil can be divided into 3 stages

1) Immediate settlement or Elastic settlement (S_e)

Compression of dry soil, moist and saturated soil without change in w

Happens immediately after load is applied

This volume change is completely reversible

Elastic settlement calculation is based on theory of elasticity

This is not a significant settlement

2) Primary consolidation settlement (S_p)

Volume change/ settlement/ compression in a **saturated soil due to the expulsion of water from the voids**

This volume change is irreversible and happens due to plastic re-arrangement

This settlement depends on the permeability of soil and the drainage condition existing in the field

Therefore, this settlement is time dependent

This settlement can be significant depending on soil and thereby resulting in the failure of structures

3) Secondary consolidation settlement (S_s)

Volume change or settlement in saturated clays after primary consolidation settlement is completed

The volume change is irreversible and happens under constant effective stress (creep behaviour)

Not as significant as S_p

Difficult to separate the end of S_p and beginning of S_s (both overlap)

Occurs mostly in highly compressive or highly organic clays

Total soil settlement

$$\text{Total settlement } St = Se + Sp + Ss$$

For sandy soils, elastic settlement is more because of better drainage (happens instantly)

For clayey soils, consolidation settlement will be significant

It is difficult to assess Sp and Ss at the time of construction

The knowledge of consolidation behavior of soil is needed to determine Sp and Ss

Consolidation behavior is determined for undisturbed soil subjected to load

Elastic or immediate settlement (Se)

$$Se = q B \left[\frac{1 - \mu^2}{E_s} \right] I_w$$

q : contact pressure intensity

B : least lateral dimension of foundation

μ : Poisson's ratio

E_s : Modulus of elasticity of soil

I_w : Influence factor (values specified for foundation)

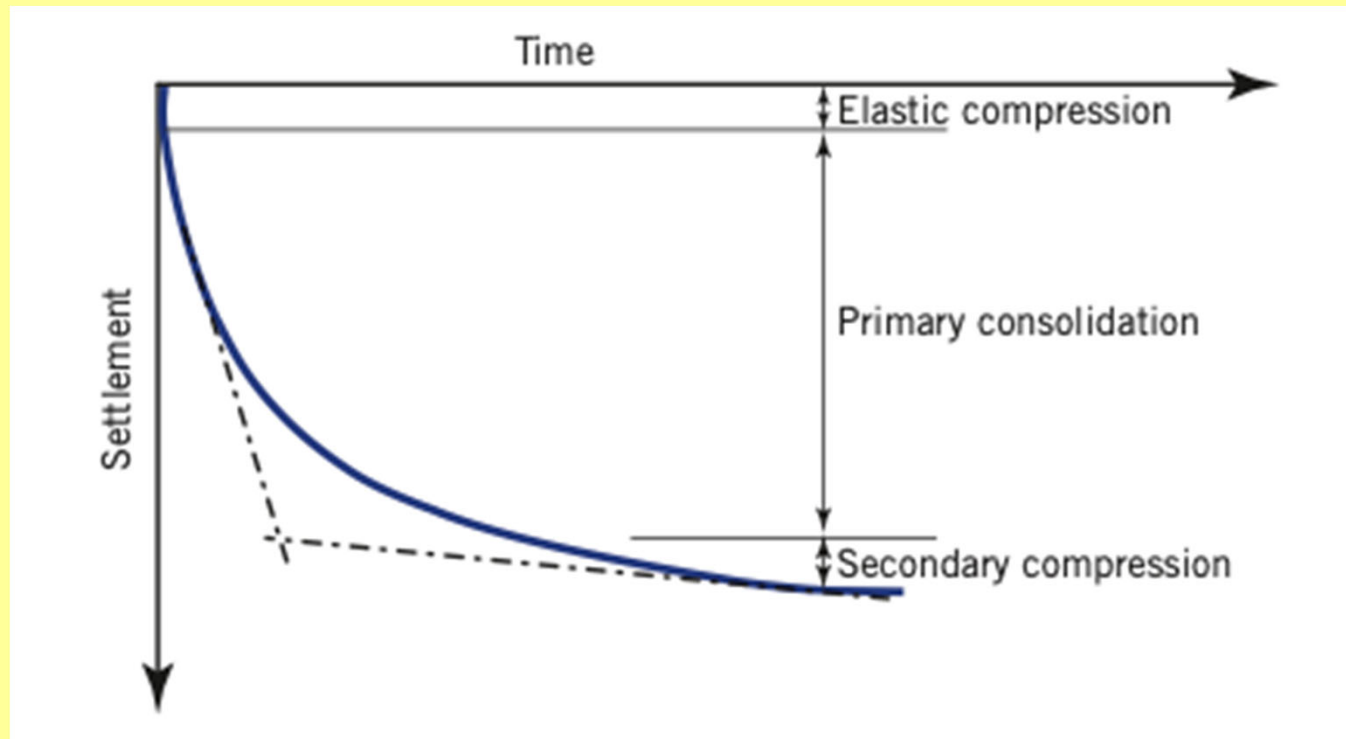
Consolidation

The decrease in volume by expelling excess pore water pressure induced by an imposed total stress

There are two aspects of consolidation behavior

- a) Compressibility of the soil (Total primary consolidation the soil undergoes)
- b) Time dependent consolidation (in what time the primary consolidation occur (rate of consolidation))

Major volume change is due to S_p



Fundamentals of consolidation

When a saturated fine grained soil is acted upon by external pressure, the pore water pressure increase instantly

The increase in u due to external load is excess pore water pressure (u_{excess})

For sandy soils, the u_{excess} is dissipated rapidly due to high permeability and immediate drainage

Therefore, sand experiences immediate settlement

When saturated clay is loaded, elastic settlement, if any, occurs instantly

Due to low permeability and poor drainage, the u_{excess} increases

Dissipation of u_{excess} through drainage takes a lot of time

Flow is laminar

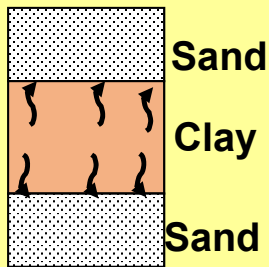
Associated consolidation settlement continues for a long period of time

Volume of water expelled is equal to the volume change in soil

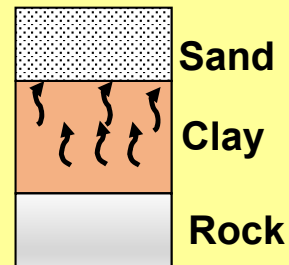
Since soil is considered as semi-infinite, change in volume is same as change in depth

Time of consolidation depends on the following factors

1. Rate of application of load (cumulative pore water pressure)
2. Coefficient of permeability of soil
3. Drainage path available i.e. one way or two way drainage



Drainage towards both sides



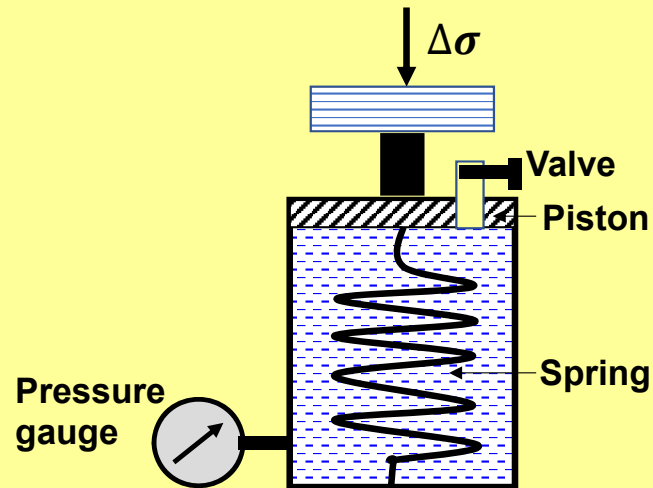
Drainage towards one side

4. Length of drainage path, which depends on thickness of soil
5. Type of drainage (manmade (PVDs) or natural)

Consolidation process : spring analogy

Spring analogy is used for explaining the consolidation process

Consider a cylinder with a spring at its centre supporting a frictionless piston



Cylinder is filled with water up to its brim and it is water tight

Valve is closed preventing any drainage (undrained condition)

External pressure is acting on the piston ($\Delta\sigma$) (piston represent foundation)

Entire pressure is borne by water instantly ($\Delta\sigma = \Delta u$) (no load transferred to spring)

Valve is opened

Water will flow out depending on valve opening (similar to k)

Flow of water occurs until the entire Δu is dissipated

$\Delta\sigma$ is borne by spring

Spring undergoes compression

Spring represents compressible soil mass

Water represents pore water

Valve represent k of soil

Stress taken by spring after dissipation of Δu represent $\Delta\sigma'$ of soil ($\Delta\sigma = \Delta\sigma'$)

Rate of piston movement downwards depends on valve opening and spring compressibility

Rate of soil settlement depends on k and soil compressibility

